FEDERAL SPECIFICATION

CLEANING COMPOUND, OPTICAL LENS

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 This specification covers one type of water-soluble, synthetic, nonionic detergent for cleaning optical lens and elements in optical shops (see 6.1).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards. The following specifications and standards, of the issues in effect on date of invitations for bids, form a part of this specification to the extent specified herein:

Federal Specification:

PPP-C-186—Containers, Packaging and Packing for Drugs, Chemicals, and Pharmaceuticals.

Federal Standards:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices listed in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Washington.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitations for bids shall apply.

American Society For Testing And Materials Standards (ASTM)

D941-55—Method of Test for Density and Specific Gravity of Liquids by Lipkin Bicapillary Pycnometer.

D1331-56—Methods of Test for Surface and Interfacial Tension of Solutions of Surface Active Agents.

D1638-59T—Method of Testing Urethane Foam Raw Materials (Hydroxyl Number).

(Copies of ASTM Standards may be obtained from the American Society for Testing and Materials, 1916 Race Street, Philadelphia 3, Pennsylvania.)

FSC 7930

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3. REQUIREMENTS

3.1 Material. The cleaning compound shall consist of a 99-100 percent active nonionic synthetic detergent.

3.1.2 *Compatibility.* No cloudiness, precipitation, or sediment shall be formed when the detergent is mixed with anionic and cationic wetting agents when tested as specified in 4.4.1.

3.2 Appearance and odor. The cleaning compound shall be a water-white to light straw colored, clear, homogeneous liquid. It shall not have an objectionable odor.

3.3 Chemical and physical properties. The cleaning compound shall conform to the requirements of table I when tested as specified in table II.

TABLE I.—*Chemical and physical properties*

Requirements	Minimum	Maximum
Water content, percent by weight	—	1.0
Ash content, percent by weight	—	0.25
Saponification number, mg. KOH/g. sample	—	5.0
Hydroxyl number, mg. KOH/g. sample	—	100
Cloud point °F. (°C.)	120°F. (48.9°C.)	160°F. (71°C.)
pH at 25°C., 1 percent solution	6.0	8.0
Surface tension at 25°C. 0.01 percent solution dynes per cm.	—	35
Specific gravity at 25°C./25°C.	1.05	1.07

3.4 Workmanship. The cleaning compound shall be a clean product free from foreign matter.

3.5 Quality. The quality shall conform to the levels established herein. The occurrence of defects shall not exceed the number permitted under the acceptable quality levels (AQL's) specified in section 4 for any examination and tests of materials, components, and end item, as applicable.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Inspection responsibility. The supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and tests. Sampling for inspection and tests shall be in accordance with the provisions of MIL-STD-105, except where otherwise indicated. For purposes of sampling, a lot shall consist of all material of the same size.

4.2.1 *Inspection of components and materials.* In accordance with 4.1 the supplier is responsible for insuring that materials and components used in the end item were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

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4.3 Inspection of the end item.

4.3.1 *Examination of the end item.* The end item shall be examined in accordance with the classification of defects in applicable sub-paragraphs below and at the inspection levels and acceptable quality levels (AQL's) set forth in 4.3.1.5. Random samples shall be drawn from each lot of end items in the same size for examination for visual, net contents and preparation for delivery defects. The lot size for purposes of determining the sample size, in accordance with MIL-STD-105, shall be expressed in units of filled primary containers of the same capacity, consisting of optical lens cleaning compound manufactured in one batch for the examination in 4.3.1.1, 4.3.1.2 and 4.3.1.3, and in units of shipping containers for examination under 4.3.1.4.

4.3.1.1 *Examination of the primary container.* The sample unit for this examination shall be one filled primary container.

Examine	Defect
Construction	Not type specified (see 5.1).
Closure	Not type specified; imperfect.
Seal	Not type specified.
Workmanship	Broken, chipped, dirty; any other defect affecting serviceability; any leaking of contents.
Markings	Missing, incomplete, incorrect, not as specified (see 5.3.1). Not affixed in place and coated with water-resistant adhesive as specified.

4.3.1.2 *Examination of contents of the primary container.* The sample unit for this examination shall be the contents of one filled primary container.

Examine	Defect
Workmanship	Not clean, contains foreign material.

4.3.1.3 *Examination of the end item for net contents.* The sample unit for this examination shall be one filled primary container. The average net contents per container shall not be less than specified or indicated for all sample units examined.

4.3.1.4 *Examination of the end item for defects in preparation for delivery.* An examination shall be made to determine that markings, materials, workmanship, weights and contents comply with the requirements of section 5 of this specification. The sample unit shall be one shipping container, fully packed, selected just prior to the closing operation. Closed shipping containers shall be examined for closure defects. The lot size shall be the number of shipping containers in the end item inspection lot.

Examine	Defect
Markings	Incorrect, incomplete, illegible, omitted; of improper size, sequence, or method of application; not in accordance with contract requirements.
Material	Any nonconforming component; component missing, damaged, or otherwise defective.
Workmanship	Inadequate application of components such as: incomplete closure of case liners, container flaps; loose or inadequate sealing, strapping, or stapling. Bulged or distorted containers.
Contents	Number of intermediate packages per shipping container is more or less than required. Number of primary containers per intermediate package is less than specified.

4.3.1.5 *Inspection levels and acceptable quality levels (AQL's) for examinations.* The inspection levels for determining the sample size and the acceptable quality levels (AQL's) expressed in defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.3.1.1	I	2.5
4.3.1.2	I	2.5
4.3.1.3	L-4	—
4.3.1.4	L-4	4.0

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TABLE II.—*Instructions for testing of the end item*

Characteristic	Specification reference		Requirements applicable to		Number determinations per unit sample	Results reported as—	
	Requirement table	Test method paragraph	Individ unit	Lot aver		Pass or fail[1]	Numerically to nearest
Water content, percent by weight	I	4.4.3	X		1		0.1
Ash content, percent by weight	I	4.4.4	X		1		0.01
Saponification number, mg. KOH/gram sample	I	4.4.5	X		1		0.1
Hydroxyl number, mg. KOH/gram sample	I	4.4.6	X		1		2
Cloud point °F. (°C.)	I	4.4.7	X		1		1
pH at 25°C.	I	4.4.8	X		1		0.1
Surface tension at 25°C., dynes per cm.	I	4.4.9	X		1		1
Specific gravity at 25°C./25°C.	I	4.4.10	X		1		0.001
Compatibility	3.1.2	4.4.1	X		1	X	
Appearance and odor	3.2	4.4.2	X		1	X	

[1] If failure is indicated, report description or numerical point of failure, as applicable.

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4.3.2 *Testing of the end item.* The end item shall be tested for the characteristics specified in table I in accordance with test methods specified in table II. The lot size shall be the number of containers of the same type filled with optical lens cleaning compound in the end item inspection lot. The sample unit shall be 1 quart of cleaning compound. The sample size shall be five. Each sample unit shall be taken from a different container. A single determination shall be made on each sample unit for all characteristics. The lot shall be unacceptable if there is any failure to meet any characteristic applicable to the sample unit.

4.4 **Test procedures.**

4.4.1 *Compatibility.*

4.4.1.1 *Preparation of sample.* One gram (g.) of the detergent shall be dissolved in 99 grams of distilled water.

4.4.1.2 *Procedure.* Mix a portion of the 1-percent solution of the detergent under test with an equal portion of a 1-percent solution of each of the following:

Sodium lauryl sulfate (anion active)

Cetyldimethylbenzyl ammonium chloride (cation active)

Record any cloudiness or precipitation.

4.4.2 *Appearance and odor.* Examine the cleaning compound for homogeneity color, clarity, presence of suspended or precipitated particles and odor.

4.4.3 *Water content.* Accurately weigh 50 grams of sample into a 250-milliliter (ml.) flask and add 100 milliliters of xylene. Connect a 25-milliliter Dean-Stark receiving tube and water condenser to the flask. Fill the tube with xylene through the top of the condenser. Heat gently on an electric hot plate to boiling and distill slowly until no more water appears to come over. Wash down inside of condenser with xylene to remove adhering drops of water. Note volume of the lower aqueous layer. Continue the distillation until the volume remains constant. Disconnect the receiving tube and allow to attain room temperature.

4.4.3.1 *Calculation.* Calculate the percent water as follows:

$$\text{Percent water} = \frac{100\ V}{W}$$

Where: V = ml. of water collected.

W = g. of sample.

4.4.4 *Ash content.* Accurately weigh 1 gram of sample and place into a weighed 30-milliliter porcelain crucible. Heat gently to volatilize any water. Gradually increase the temperature, being careful to avoid boiling over and spattering. Finally, ignite to constant weight at 600°C. Cool in a desiccator and weigh.

4.4.4.1 *Calculation.* Calculate percent ash as follows:

$$\text{Percent ash} = \frac{100\ A}{W}$$

Where: A = g. of residue.

W = g. of sample.

4.4.5 *Saponification number.* Accurately weigh approximately 5 grams of the sample into a 300-milliliter Erlenmyer flask and add from a pipette 50.0 milliliters of approximately half-normal alcoholic potassium hydroxide. Heat on a water or steam bath under reflux for 1 hour. Add three drops of 1 percent phenolphthalein in alcohol and titrate with a standardized half-normal sulfuric acid. Repeat the procedure on a blank.

4.4.5.1 *Calculation.* Calculate the saponification number as the number of milligrams

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of potassium hydroxide required for 1 gram of the sample as follows:

$$\text{Saponification number} = \frac{(A - B) \times N \times 56.1}{W}$$

Where: A = ml. of standard H_2SO_4 required for blank titration.

B = ml. of standard H_2SO_4 required for sample titration.

N = normality of sulfuric acid.

W = sample weight in grams.

4.4.6 *Hydroxyl number.* The hydroxyl number shall be determined as specified in ASTM Method D1638-59T.

4.4.7 *Cloud point.* Dissolve 0.5 grams of sample in 100 milliliters of distilled water. Transfer 10 milliliters of this solution to a clean, dry, 25-by 200-mm. test tube, and insert a thermometer. The test tube shall then be inserted about half-way in an 800-milliliter beaker nearly full of water. Mechanical stirring of the water shall be provided and low heat shall be applied to raise the temperature slowly. The sample shall not be stirred until a preliminary indication of the cloud point is obtained by turbidity developing in the layers next to the wall of the test tube. From the time this is observed until the solution suddenly becomes cloudy, the sample shall be stirred continuously by the thermometer. The cloud point shall be reported in degrees Fahrenheit (°F.) as the temperature at which cloudiness appears throughout.

4.4.8 *pH value.* Dissolve 1.0 grams of the sample in freshly boiled and cooled distilled water, dilute to 100 milliliters with freshly boiled and cooled distilled water, and mix. Determine the pH value (to the nearest 0.1) of this solution at 25°C. by means of a suitable pH meter employing a glass electrode.

4.4.9 *Surface tension.* Dissolve 0.1 gram of sample in freshly boiled and cooled distilled water, dilute to 100 milliliters and mix. Determine the surface tension of this solution at 25°C. in accordance with ASTM Method D1331-56.

4.4.10 *Specific gravity.* Determine the specific gravity of the cleaning compound at 25°C/25°C. in accordance with ASTM Method D941-55.

5. PREPARATION FOR DELIVERY

(Fed. Std. No. 102 shall be referred to for definitions and application of the various levels of packaging and packing for civil agency procurements.)

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 *Level A.*

5.1.1.1 *Unit packaging.* Unit packaging shall be one pint or one quart, as specified (see 6.2). Containers shall conform to group A, class 1, type e, style 1, grade 1 or 2, closure A, seal A or B of PPP-C-186.

5.1.1.2 *Intermediate packaging.* Twelve pint- or twelve quart-containers shall be intermediately packaged in accordance with the intermediate packaging requirements of PPP-C-186.

5.1.2 *Level C.* The cleaning compound shall be packaged in accordance with the industry's practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 *Level A.* Cleaning compound, packaged as specified in 5.1, shall be packed in accordance with the level A requirements as specified in PPP-C-186.

5.2.2 *Level B.* Cleaning compound, packaged as specified in 5.1, shall be packed in accordance with the level B requirements as specified in PPP-C-186.

5.2.3 *Level C.* Cleaning compound, packaged as specified in 5.1, shall be packed in

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a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with rules or regulations of carriers applicable to the mode of transportation.

5.3 **Marking.**

5.3.1 *Labeling.* Label marking shall be placed on a label made of white paper as specified in MIL-STD-129. Unless otherwise specified, each container of cleaning compound shall be durably and legibly labeled with the following information:

CLEANING COMPOUND, OPTICAL LENS

Preparation of Cleaning Solution

A solution suitable for cleaning glass and plastic optical lens and other optical elements not affected by water shall be prepared by dissolving 1 ounce of this cleaning compound in a small amount of warm water and making up to 1 gallon with distilled or clear fresh water. Best results will be obtained by adding the cleaning compound in small amounts to the water with vigorous stirring.

CLEANING PROCEDURE

1. Rinse the lens in clean running water to wet surface and remove particles that might scratch the polished surface.

2. Apply the prepared solution to the surface to be cleaned using a wad of absorbent cotton or a soft clean cloth, free from sizing.

3. Rinse the surfaces in distilled water.

4. Repeat steps 2 and 3 to remove stubborn soils.

5.3.2 *Civil agencies.* In addition to labeling specified in 5.3.1 and any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.3 *Military requirements.* In addition to labeling specified in 5.3.1 and any special marking required by the contract or order, interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 **Intended use.** The cleaning compound covered by this specification is intended for use in cleaning plastic and glass lenses, coated or uncoated, and other optical elements. Its principal use shall be in optical repair and maintenance shops.

6.2 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Inspection responsibility, methods, and tests (see section 4).

(*c*) Selection of the applicable levels of packaging and packing (see 5.1 and 5.2).

(*d*) Pint or quart size required (see 5.1).

6.3 **Standard pack (civil agencies).** The Standard Pack for this commodity shall be as specified in 5.1.1 and 5.2.2. The Standard Pack requirements are intended for use in procurements of stores stock replenishments. Procuring officers should use the Standard Pack requirements when it is known that the material will be shipped from a supplier to a domestic warehouse, supply depot, or immediate storage point for temporary storage, subsequent issue, or shipment to eventual user.

6.4 **Transportation description.** Transportation descriptions and minimum weights applicable to this commodity are:

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Rail:

Cleaning Compound, not otherwise indexed by name, liquid.

Carload minimum weight 36,000 pounds.

Motor:

Cleaning Compound, not otherwise indexed, liquid.

Truckload minimum weight 36,000 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY CUSTODIANS:

Army—GL

Navy—Sh

Air Force—MOA

OTHER INTEREST:

Army—M

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